1 Introduction

In this report, I will use the Metropolis algorithm (a Markov chain Monte Carlo method) to simulate a 2D Ising model. I will begin by simulating a 16 x 16 square lattice of spins for 20 different temperatures, plotting relevant average quantities for each temperature. I will then investigate the critical behaviour of the simulation. To do this, I will first look at how the specific heat and magnetic susceptibility change with lattice size. Then, I will look at how the simulation behaves at temperatures near the critical temperature, which is the temperature at which a phase transition is expected.

2 Single Lattice

In order to simulate a 2D Ising model, I first initialised the lattice of spins such that all spins were up. Next, we use a random number generator to randomly select one of the spins in the lattice. This spin is then flipped, and the change in energy, ΔE, associated with the flip is calculated. If $\Delta E \leq 0$, we accept the flip. If $\Delta E > 0$, then we generate a uniform random number r, between 0 and 1. Then, if $r \leq \exp(-\Delta E/k_B T)$, we accept the flip, otherwise, we reject the flip, and return the spin to its previous state (ie, we 'unflip' the spin).

In my simulation, I performed this process for 1100 sweeps of the lattice, where one sweep corresponds to performing N_s flips (N_s = the number of spins in the lattice). After every sweep, I calculated relevant values, including the energy, magnetisation, specific heat, and magnetic susceptibility. I then removed the first 100 values. This is because for the first 100 sweeps, the system is not in thermodynamic equilibrium, so including values calculated during this time would result in an inaccurate average. After removing the first 100 values, I determined the average of the remaining values.

I performed this process for 20 temperatures between $T = 0$ and $T = 5$. Instead of reinitialising the lattice for every temperature, I used the final lattice for a given temperature as the initial lattice for the next temperature. This meant that it took less time for the system to reach thermodynamic equilibrium, especially for higher temperatures.

A plot of the average energy per spin against temperature is shown in figure 1. To determine the average energy, first the initial energy of the system was calculated, using the formula:

$$E = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - \mu B \sum_{i}^{N_s} \sigma_i \tag{1}$$

Then, after each flip, the change in energy

$$\Delta E = -2\sigma_{\text{flipped}} \left[J \left(\sigma_{\text{left}} + \sigma_{\text{right}} + \sigma_{\text{above}} + \sigma_{\text{below}} \right) + \mu B \right] \tag{2}$$

was added to the total energy. Here, σ_{flipped} is the value of the flipped spin after it has been flipped. The total energy was recorded for each sweep and divided by the number of spins to get the total energy per spin. The average of these values was then taken to get the average energy per spin. The error in these values was calculated using the equation

$$E_{\text{error}} = \sqrt{\frac{\langle E^2 \rangle - \langle E \rangle^2}{N_{\text{sweeps}}}} \tag{3}$$

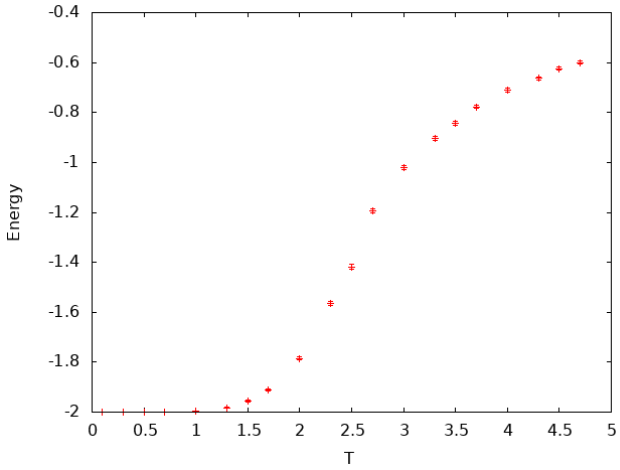


Figure 1: Average energy per spin

The average magnetisation per spin was calculated in a similar way to the average energy per spin. First, the initial magnetisation was calculated using the formula

$$M = \sum_{i}^{N_s} \sigma_i \tag{4}$$

Then, after each flip, the change in magnetisation

$$\Delta M = 2\sigma_{\text{flipped}} \tag{5}$$

was added to the total magnetisation. The total magnetisation was recorded for each sweep, and divided by the total number of spins. The average of these values was then taken to get the average magnetisation per spin. The error in the average magnetisation per spin was calculated using the formula

$$M_{\text{error}} = \sqrt{\frac{\langle M^2 \rangle - \langle M \rangle^2}{N_{\text{sweeps}}}} \tag{6}$$

A plot of the average magnetisation per spin against temperature is shown in figure 2. This plot also shows the analytic magnetisation for comparison. The computational magnetisation matches the analytic values closely for small temperatures, but diverges for temperatures above the critical temperature $T \approx 2.27$.

The specific heat was determined using the equation

$$c = \frac{\left(\langle E^2 \rangle - \langle E \rangle^2\right) N_s}{k_B T^2} \tag{7}$$

A plot of this is shown in figure 3. This figure shows that the specific heat has a peak at the critical temperature.

The magnetic susceptibility per spin was calculated by

$$\chi = \frac{\left(\langle |M|^2 \rangle - \langle |M| \rangle^2\right) N_s}{k_B T} \tag{8}$$

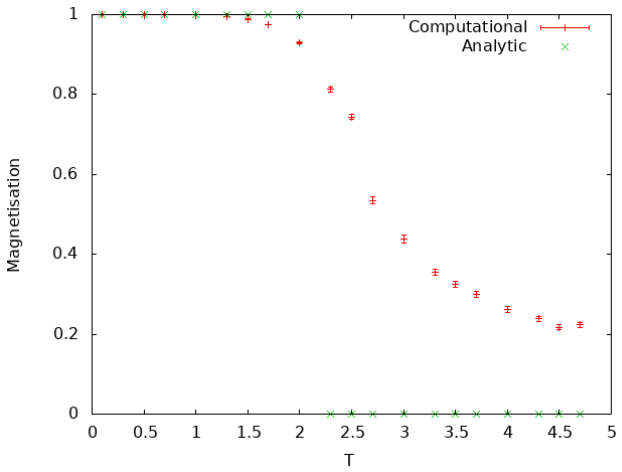


Figure 2: Average magnetisation per spin

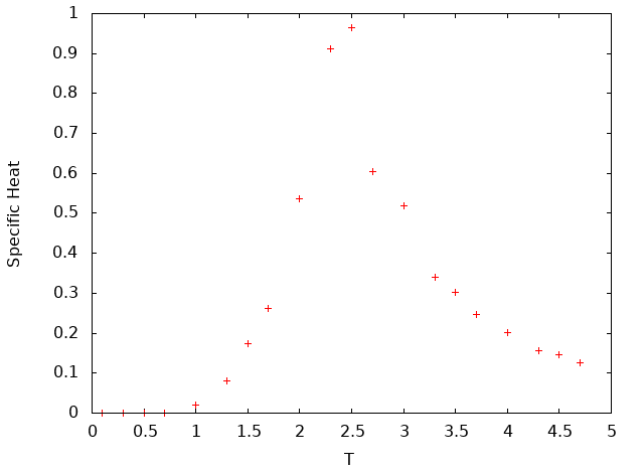


Figure 3: Specific heat

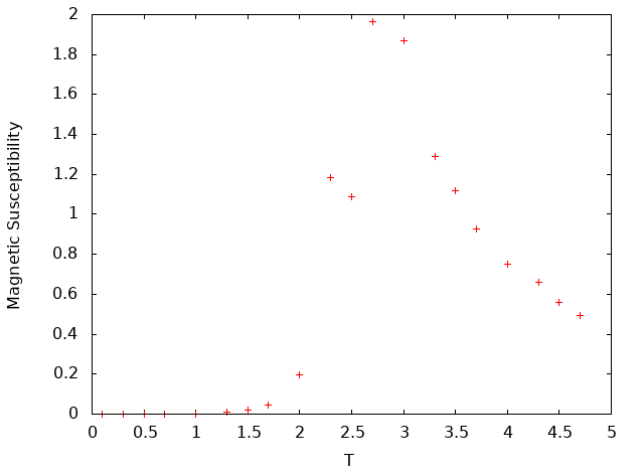


Figure 4: Magnetic susceptibility

A plot of the magnetic susceptibility per spin is shown in figure 4. Similar to the specific heat, the magnetic susceptibility also has a peak at the critical temperature.

3 Varying Lattice Size

In order to observe the effect of lattice size on the specific heat and magnetic susceptibility, we can run the simulation for four lattice sizes: $L = 8, 16, 32$, and 64. Figures 5 and 6 show the heat capacity and magnetic susceptibility, respectively, as a function of temperature for all four lattice sizes. In both these plots, the peak at the critical temperature is higher for larger lattices. This suggests that as the lattice size increases, we would expect both the specific heat and magnetic susceptibility to go to infinity at the critical temperature.

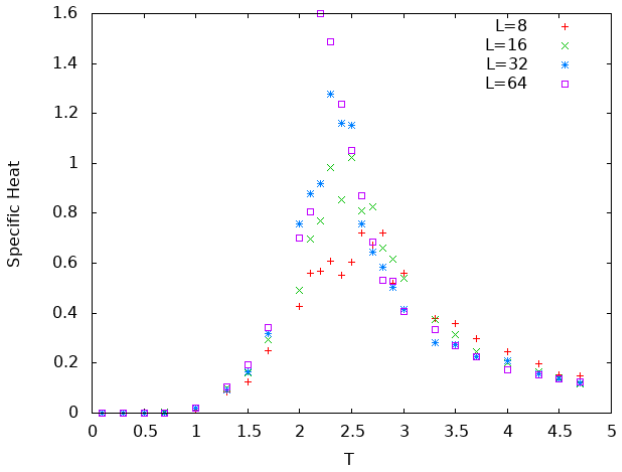


Figure 5: Specific heat

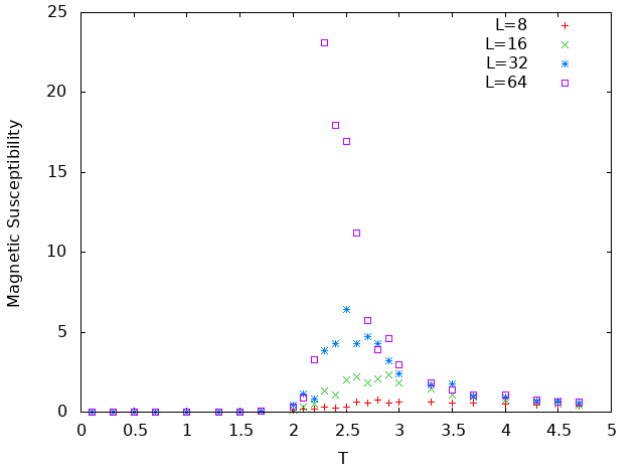


Figure 6: Magnetic susceptibility

The specific heat and magnetic susceptibility should obey finite-size scaling laws:

$$c(t) = \log(L) f\left(L^{\frac{1}{\nu}} t\right) \qquad \chi(t) = L^{\frac{\gamma}{\nu}} g\left(L^{\frac{1}{\nu}} t\right) \tag{9}$$

where f and g are functions that are independent of the lattice size, and the parameters are $\nu = 1$ and $\gamma = 7/4$ for the 2D Ising model. To check that my simulation obeys these scaling laws, we can check that the curves for each of the four lattice sizes collapse to a single curve,

when the axes are rescaled according to equation 9. To do this, I plotted $c(t)/\log(L)$ against $L^{1/\nu}t$ and $\chi(t)/L^{\gamma/\nu}$ and $L^{1/\nu}t$.

Figures 7 and 8 show the rescaled specific heat and magnetic susceptibility, respectively. At a first glance, the four curves in both plots appear not to collapse very well. However, the y-axis is scaled much smaller than the x-axis. Therefore, the vertical distance between points on this plot is actually very small. Hence, around $t = 0$, the curves on both plots do roughly collapse to a single curve, as expected.

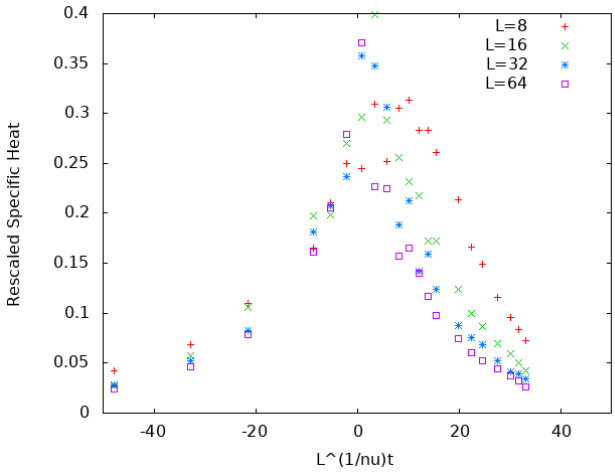


Figure 7: Rescaled specific heat

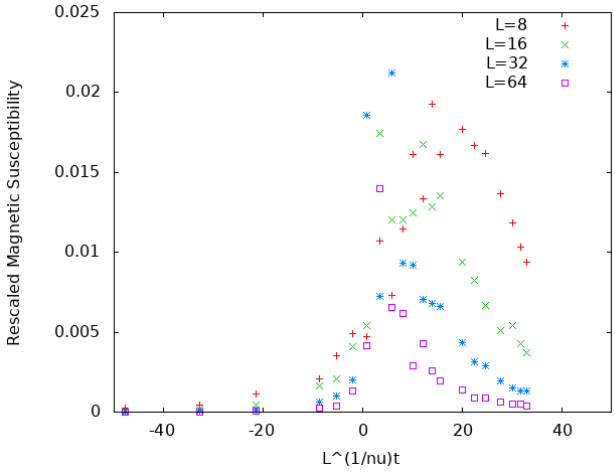


Figure 8: Rescaled magnetic susceptibility

4 Critical Region

In the thermodynamic limit, the magnetisation per spin and magnetic susceptibility should scale as:

$$M \approx |T - T_c|^{\beta} \qquad \chi \approx |T - T_c|^{-\gamma} \tag{10}$$

where $\beta = 1/8$ and $\gamma = 7/4$. To check that this applies for my simulation, I ran the simulation for 13 temperatures between $T = 2$ and $T = T_c$, for a lattice with $L = 64$, looping over 10000 sweeps. The large sweep number and lattice size is to ensure accurate results. Plots of equations 10 on a log-log scale are shown in figures 9 and 10. Both of these plots are roughly linear, as expected.

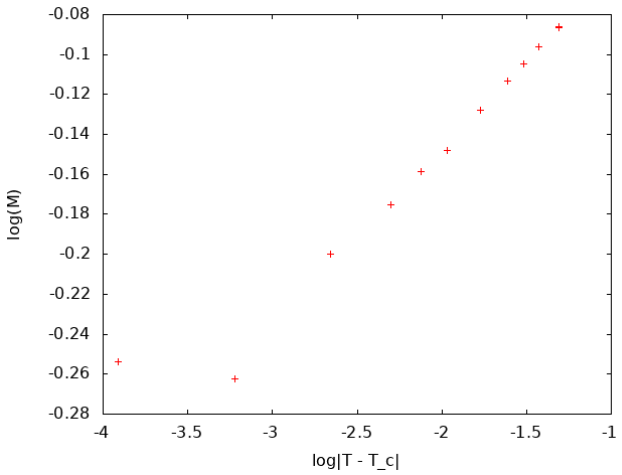


Figure 9: Magnetisation power law scaling

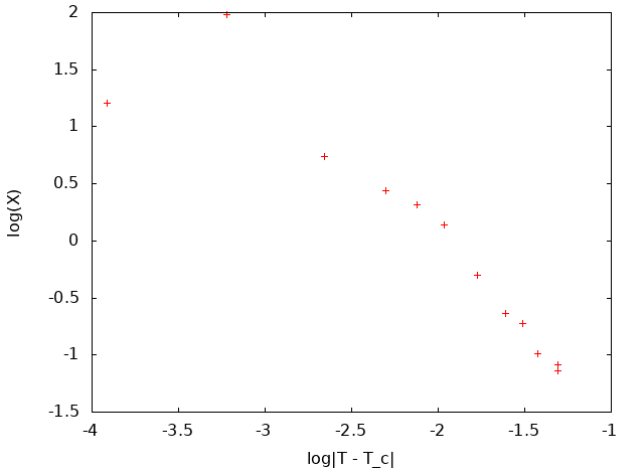


Figure 10: Magnetic susceptibility power law scaling

5 Conclusion

In conclusion, I found that for my 2D Ising model, the magnetisation per spin closely matched the analytic values for small temperatures, but diverged for temperatures larger than the critical temperature. Furthermore, the specific heat and magnetic susceptibility both had a peak at the critical temperature. We also saw that the these peaks were higher for larger lattices. My simulation obeyed the finite-size scaling laws, as expected. Furthermore, the magnetisation and magnetic susceptibility scaled as power laws around the critical temperature.